The attention mechanism can be plainly written as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

The cross-attention mechanism can be written as:

$$\begin{aligned}&\text{Cross Attention}(X_q, X_c; W_q, W_k, W_v)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)(X_cW_v)\end{aligned}$$

The relational-attention mechanism can be written as:

$$\begin{aligned}&\text{Relational Attention}(X_q, X_c; W_q, W_k, V)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)V\end{aligned}$$

The adaptable-attention mechanism can be written as:

$$\begin{aligned}&\text{Adaptable Attention}(X_q, X_c; \gamma, W_q, W_k, W_v)\\ &\quad= \text{softmax}\left(\frac{(X_qW_q)(X_cW_k)^\top}{\sqrt{d}}\right)\left[(1-\gamma)X_c + \gamma I\right]W_v\end{aligned}$$

The pairwise transitive inference task can be described by:

Obs.?	x_i	x_j	$x_i > x_j$
Yes	A	B	1
	B	C	1
	B	A	0
	C	B	0
No	A	C	1
	C	A	0

Table 1: Truth table for transitive inference.